0073. Set Matrix Zeroes

Medium · Math & Geometry

leetcode.com/problems/set-matrix-zeroes

Problem

- If an element is 0, set its entire row and column to 0, in place. Follow-up: O(1) extra space.

Intuition

- Use the first row and first column as marker arrays. Two booleans remember whether the first row and column themselves must be zeroed.

Brute Force

- Copy the matrix: O(mn) space. Row and column boolean arrays: O(m + n) space.

Optimal Approach

- FirstRow / firstCol = whether they contain a zero.
- For r, c >= 1: if m[r][c] == 0, set m[r][0] = m[0][c] = 0.
- For r, c >= 1: zero the cell if m[r][0] == 0 or m[0][c] == 0.
- Zero the first row and column if their flags are set.

Worked Example

```
[[0,1,2,0],[3,4,5,2],[1,3,1,5]].

firstRow has a zero -> true ; firstCol has a zero -> true
inner cells in columns 0 and 3 -> 0
finally zero row 0 and column 0
result [[0,0,0,0],[0,4,5,0],[0,3,1,0]]
```

Java Solution

```java
public class SetMatrixZeroes {
    public void setZeroes(int[][] m) {
        int rows = m.length, cols = m[0].length;
        boolean firstRow = false, firstCol = false;
        for (int c = 0; c < cols; c++) if (m[0][c] == 0) firstRow = true;
        for (int r = 0; r < rows; r++) if (m[r][0] == 0) firstCol = true;
        for (int r = 1; r < rows; r++)
            for (int c = 1; c < cols; c++)
                if (m[r][c] == 0) { m[r][0] = 0; m[0][c] = 0; }
        for (int r = 1; r < rows; r++)
            for (int c = 1; c < cols; c++)
                if (m[r][0] == 0 || m[0][c] == 0) m[r][c] = 0;
        if (firstRow) for (int c = 0; c < cols; c++) m[0][c] = 0;
        if (firstCol) for (int r = 0; r < rows; r++) m[r][0] = 0;
    }
}
```

Complexity

- Time: O(m n). Space: O(1).

Pitfalls

- Process the first row and column last, or the markers overwrite real data too early.